

Radhika Gaonkar

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Summary

Researcher at Prime Intellect working on post-training and reinforcement learning for self-improving, long-horizon agents. Research focuses on agentic RL, synthetic data and environments, rewards/verifiers, trajectory-level evaluation, and reward hacking. Previously Tech Lead at Microsoft 365 Copilot, leading post-training and evaluation for reasoning, domain adaptation, and personalization systems deployed at enterprise scale.

Technical Skills

- **Post-Training & Alignment:** RLVR · GRPO · DPO · PPO · On-Policy Distillation · Reward Modeling · SFT · Mid-Training
- **LLM Evaluation:** LLM-as-a-Judge · Agent-as-a-Judge · Agentic Evaluation · Uncertainty Estimation & Calibration
- **Data & Systems:** Synthetic Data/Environments · PyTorch · Transformers · prime-RL · vLLM · Ray

Experience

Member of Technical Staff, Prime Intellect - RL & Agents

Jul 2026 - Present

- Agentic RL & self-improving agents: Research on post-training, synthetic environments, and benchmarks for robust long-horizon agents, with emphasis on reliable learning signals, reward hacking behavior and continual learning.
- Core contributor to post-training open-weight models and building new benchmarks for improving reasoning capabilities on enterprise workflows, as part of the Nemotron Coalition with Nvidia.

Senior Applied Scientist, Microsoft - M365 Copilot Research

Post-Training & Reasoning Models

March 2025 - Jul 2026

- **~85% CoT reduction, 50% latency drop, 97% accuracy, shipped worldwide:** Led two-phase RL post-training recipe (on-policy distillation and RLVR) to adapt [Copilot](#)'s reasoning model to internalize query difficulty, eliminating an external routing classifier; hard-task reasoning quality maintained across agentic benchmarks.
- **92% task success, 68% OOD generalization:** Post-training lead for multi-turn enterprise optimization agent ([Copilot Tuning](#)), which solves complex enterprise queries as an optimization problem: RLVR with solver-as-verifier, synthetic data generation, and LLM-as-a-Judge evaluator across multiple enterprise domains.
- **Agent-as-a-Judge grader adopted across Copilot RL pipelines:** Led development of agent-loop-based evaluator, replacing static LLM-as-judge with tool-grounded scoring; integrated into RLVR training pipeline; ~82% correlation with production eval; drove cross-team adoption.

Domain Adaptation, Personalization & Synthetic Data

Jan 2024 - Feb 2025

- 100x data scaling, ~88% factuality: Led synthetic-data research for domain-expert SLMs spanning mid-training, post-training, and evals; shipped as specialized sub-agents in Researcher.
- +8.5% thumbs-up / +22% style similarity / +21% MAU: Led mid-training, SFT/DPO, sparse-evidence personalization, and memory evaluation improvements for Copilot.

LLM Routing, Fine-Tuning & Model Efficiency

Feb 2020 - Dec 2023

- 60% eval cost reduction, 27% latency drop; 23% retrieval recall gain; 3% CTR lift, 35% latency reduction: Built production LLM routing, user embeddings, LoRA fine-tuning, and model-compression systems for M365.

Machine Learning Scientist, Fractal AI

May 2016 - Dec 2017

- Deployed NLP and CV systems (sentiment analysis, topic modeling, OCR, object detection) for F500 clients (Visa, P&G, StateFarm, Google); led model selection, training, and production deployment.

Research Assistant, Technische Universität Darmstadt

Jan 2014 - April 2016

- Research on argument mining, SocialNLP, and [Reinforcement Learning - based](#) recommender systems at [UKP Lab](#).

Education

- Master of Science, Computer Science at Stony Brook University, New York 2018 - 2019
- Bachelor of Engineering (Honors), Computer Science at BITS Pilani, India 2010 - 2014

Select Publications and Talks

- TRACE: Diagnosing Verifier Brittleness in Agentic Evaluation - **COLM AIMS Workshop 2026**
- Reward Stress Tests for Agentic Post-Training with Ray - **Ray Summit 2026**
- Your Agent Aced the Eval, Failed the User (poster) - [Agentic AI Summit 2026](#)
- Building Agentic Evaluators for Enterprise LLM Systems (talk) - [Women Who Code Summit 2026](#)
- Permissive Information-flow analysis for LLMs; Siddiqui, **Gaonkar**, Köpf et al. - [TMLR 2025](#)
- Knowledge Infused Decoding; Liu, Zheng, Gupta, **Gaonkar** et al. - [ICLR 2022](#)
- Modeling Label Semantics for Predicting Emotional Reactions; **Gaonkar**, Kwon, Bastan et al. - [ACL 2020](#)